



Mapping Cognition to Software Modelling & Testing

Alan Richardson & Gustav Khun
@EvilTester & @GustavKuhn

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Blurb

Have you ever wondered how studying magic can improve your software testing? Alan Richardson talked his way into the Magic Circle in London, allowing him to learn how to reverse engineer the process of observing the performance of magic and develop techniques which he applied directly to improve his Technical Testing process.

Have you ever wondered why magic works? Gustav Kuhn heads the MAGIC lab at Goldsmiths University of London, and his his research lab uses magic to study a wide range questions about human cognition. For example, why is your mind so easily tricked? How can magicians misdirect your attention, and why is your experience distorted by perceptual, attentional and other more general cognitive illusions?

Welcome

- Who we are
- What we will be doing
- Logistics 1.5 hours

Beginnings



Hieronymos Bosch - The Conjurer
@EvilTester | @GustavKuhn

Don't ask where the Lemons come from

Magic and Psychology

- Why I became a psychologist
- The Science of Magic

Magic and Testing

- “I talked my way into the Magic circle”
 - The library
 - The Theatre
- I learned how to observe magic
 - Watch Magic
 - Hypothesise
 - Study Domain - to learn what to look for

Magic and HCI - The user illusion

The Mechanical Turk

- History of the Mechanical Turk
- Impact on Society and Technology - Machines challenge human intellect / inspiration for AI
- Is Turk Real or Trick?
- How would you test it?

Principles

Principle in magic: Give people the illusion they are testing the situation, but magician is always in control of the situation.

- I asked you to examine/test the cups, but you cannot test the rest of the trick
 - Does not work all the time (e.g. burned down theatre, switched bullet in bullet catch)
- Magicians never repeat the same trick

The User Illusion - Shared virtual realities

- Illusions



You will choose...

You will choose...

Ace of Hearts

Ten of Spades

Marco Tempest - Cyber Magic Card Trick

- Augmented Reality Magic
- [Marco Tempest](#)
- A Cyber Magic Card Trick Like No other
- youtube.com/watch?v=ZX8MBBohX3s
- [Ted Talks Listing](#)

How Magic Works

- prevent you from attributing the true cause of an effect

Illusory Attribution

- Illusory Attribution - Michotte Causal attribution video
 - We are willing to endorse illusory causal attributions
 - <https://www.youtube.com/watch?v=VTNmLt7QX8E>

Magic relies on exploiting limitation in cognition

Magic relies on exploiting limitation in cognition

- People are willing to endorse the illusory causal attribution (i.e. the magic stuff) because they are not aware of their Cognitive limitations and Biases

Surprising Limitations and Cognitive Biases

- Limitations in perceptual awareness (misdirection trick)
- Subjective nature of visual experience (seeing is believing) - card trick
- Biases in reasoning (Theory of false solution)

“How Magic Works” Impacts How We Test

- Understanding magic:
 - Learn how to observe differently
 - Contextual understanding of magician’s techniques
 - Gimmicks and new technologies
 - Multiple Methods & Effects

How Magic Works Impacts How We Test

- Not take what we see as absolute
 - Explanations are possible not definite
- Take action to help us model reality
 - Take notes,
 - Perform experiments, Learn to falsify
- Communicate
 - It seemed as though...
 - It looked like....
 - E-Prime !be

“How Magic Works” Impacts How I Test

- Modelling
- Observation
- Reflection
- Interrogation
- Manipulation

Modelling

Modelling Systems

E.g. Testing a web application

- Domain knowledge
 - HTTP, Cookies, JavaScript
 - More knowledge, deeper model
- Surface Interaction e.g. HTML rendering
 - Effect
- GET, PUT, POST, Ajax
 - Method

Our model is limited by our knowledge.

Observation

- we need to learn how to observe, this is contextual
 - i.e. if it is a web app we learn to observe the HTTP messages, or the DOM changing, or the javascript events
 - if it is Magic then we learn to observe hand movements, what we are directed to, what we are directed away from, why did they do that?

Observation

- supports incrementally building a model which only becomes realised after we have finished and reflect on it
 - Note taking as we go

We see less than we think we do

- Attentional misdirection
- Change Blindness
- Inattentional Blindness
 - HCI display in pilots - Radiographers
- Never perform the same trick twice
- Don't get distracted
- Multitasking myth

Reflection

- think about the model
- what are the component parts, what are our assumptions?
- what to test next, for what purpose?

Interrogation

- when you first see a magic trick you don't necessarily know what will happen at the end so you have to go back and interrogate to validate your model against what you have modelled
- you interrogate to see where you were led astray during your incremental model building
 - Look at the observed results in more detail
 - Tools can create logs to review later

Manipulation

- “If in doubt, try it out.”
- is your model or hypothesis actually possible?

In magic you never give people the opportunity to test it.

Summary

Gustav Kuhn

- <https://www.gold.ac.uk/psychology/staff/kuhn/>
- <https://twitter.com/gustavkuhn>
- <https://scienceofmagicassoc.org/>

Alan Richardson

Alan Richardson www.compendiumdev.co.uk

- LinkedIn - [@eviltester](#)
- Twitter - [@eviltester](#)
- Instagram - [@eviltester](#)
- Facebook - [@eviltester](#)
- Youtube - [EvilTesterVideos](#)
- Pinterest - [@eviltester](#)
- Github - [@eviltester](#)
- Slideshare - [@eviltester](#)

Alan Bio

Alan is a Software Development and Testing Coach/Consultant who enjoys testing at a technical level using techniques from psychotherapy and computer science. In his spare time Alan is currently programming a [Twitter client called ChatterScan](#), and [multi-user text adventure game](#). Alan is the author of the books "[Dear Evil Tester](#)", "[Java For Testers](#)" and "[Automating and Testing a REST API](#)". Alan's main website is compendiumdev.co.uk and he blogs at blog.eviltester.com